Full fine-tuning versus LoRA

Correct comparison: pretrained GPT-2 (reference), full fine-tuning on the new data, and LoRA fine-tuning on the same data. Full fine-tuning is now trained and evaluated for the same budget as LoRA; the previous 40-step full-training run was a memory-only baseline.

Controls: same GPT-2 starting weights, training examples and sample order, FP32 with TF32 off, batch 8, padded length 128, 1,200 steps, seeds 42/43/44, and NVIDIA A30. Starting-weight and batch-order SHA256 fingerprints match between methods for every seed. Each run processes 1,228,800 padded token positions; supervised answer-token budgets also match by seed.

Equal tuning budget: both methods tried learning rates 0.001 and 0.0001 for 1,200 steps on seed 42. Validation NLL at steps 300/600/900/1200 selected the rate and checkpoint; both rate choices were saved before new test evaluation. Selected rates: full 0.0001, LoRA 0.001. Seeds 43/44 then used the chosen rate. Four training runs per method including the unselected candidate; six selected runs in the final comparison.

Data: 480 train matches, 40 validation matches, 40 test matches, three questions per match, plus 120 score-swapped test questions. Test team names and score combinations are absent from training. This is the existing disclosed synthetic benchmark, not a newly blinded test set; no new test-based tuning was used.

Primary-answer score: pretrained 25.8%; full 31.9 ± 14.4%; LoRA 81.7 ± 0.8%. Answer + score fields: pretrained 0.0%; full 31.9 ± 14.4%; LoRA 80.8 ± 1.4%. ± is sample SD across three training seeds, not a confidence interval.

Parameter and runtime efficiency

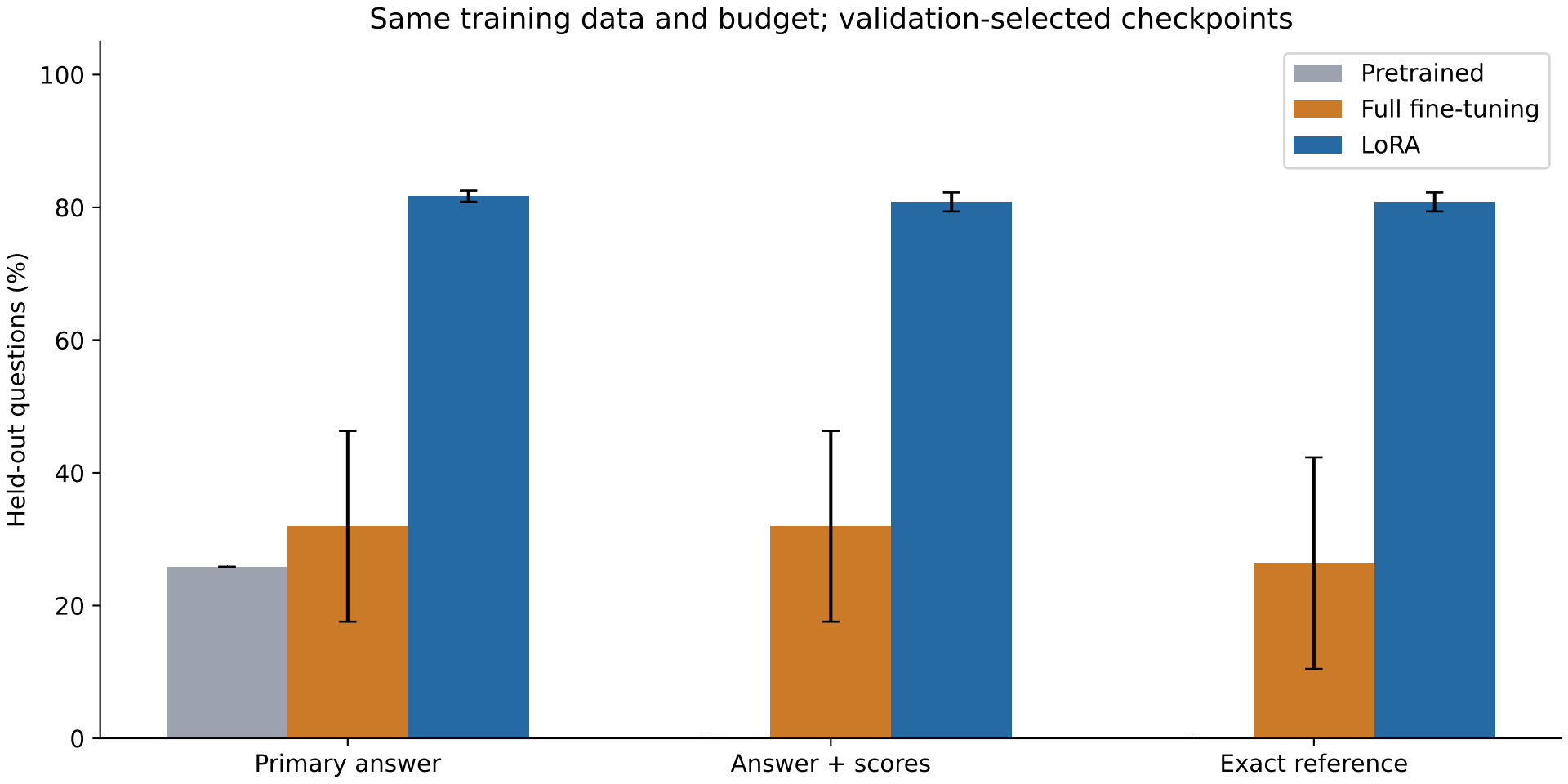
Trainable scalar parameters: full 163,037,184 (100%); LoRA 589,824 (0.3605% of the LoRA model). LoRA total 163,627,008; full total 163,037,184. LoRA updates 276.4 times fewer trainable parameters. All LoRA base-weight audits passed.

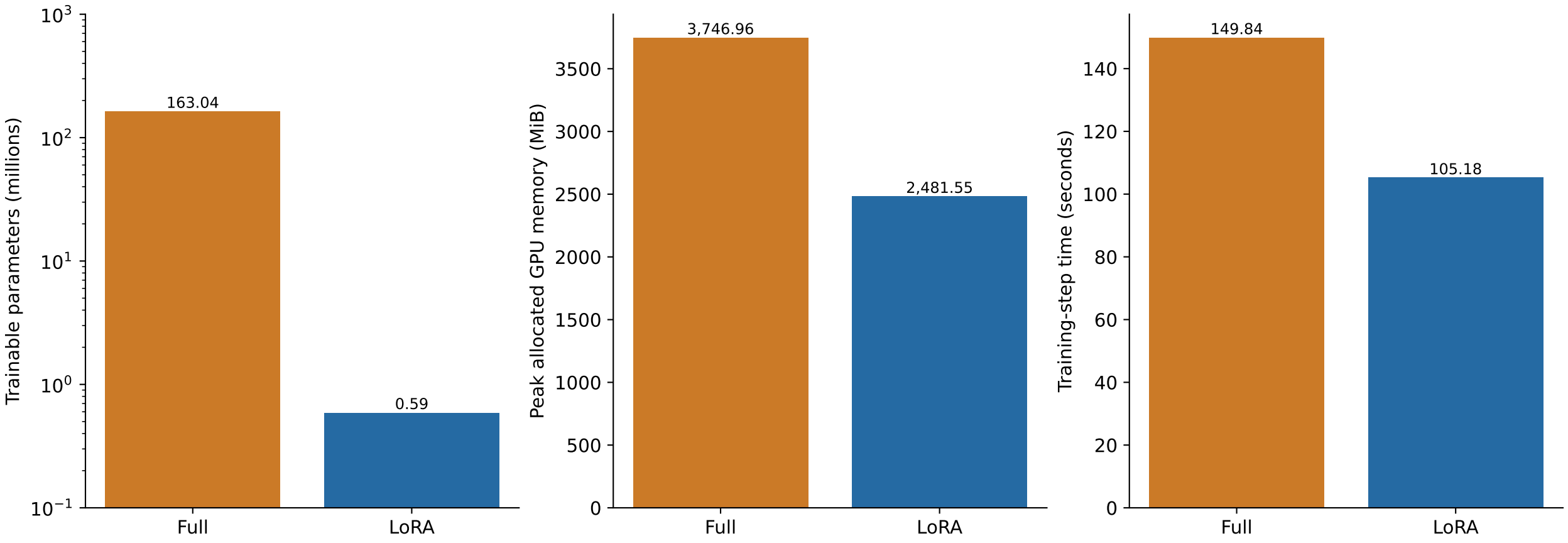
Mean full-budget peak allocated GPU memory: full 3747.0 MiB; LoRA 2481.6 MiB (33.8% lower). Mean synchronized training-step time: full 149.8 s; LoRA 105.2 s. Timing excludes transfer, validation, audits, checkpoint I/O and test generation. End-to-end training wall times are reported separately in JSON.

Limits: metrics are transparent but format-sensitive field checks, not broad conversational quality. Three questions share each match. Generated templates can be reused, so score-swap consistency and failures matter. Both methods are trained for the full budget but evaluated at their validation-selected checkpoint. This does not establish quality equivalence or time-to-quality without further experiments.

FP32 parameter storage is 4 bytes per scalar. Allocated and reserved GPU memory are separate, non-additive allocator measurements; peaks include activations, gradients and optimizer states. They exclude some driver/context memory. minGPT stores separate input/output embedding matrices, so its physical parameter count exceeds tied GPT-2.

The older reports compared LoRA quality to an untouched reference and used a 40-step full-training memory run. Use this matched report for full-versus-LoRA claims.


Same training data and budget; validation-selected checkpoints
Held-out questions (%)
100
80
60
40
20
0
Pretrained
Full fine-tuning
LoRA
Primary answer
Answer + scores
Exact reference


Trainable parameters (millions)
10^3
10^2
10^1
10^0
10^{-1}
163.04
0.59
Full
LoRA
Peak allocated GPU memory (MiB)
0
500
1000
1500
2000
2500
3000
3500
3,746.96
2,481.55
Full
LoRA
Training-step time (seconds)
0
20
40
60
80
100
120
140
149.84
105.18
Full
LoRA

First shared exact-reference match

Basketball match: Eagles vs Sharks. Halftime score: Eagles 41, Sharks 28. Final score: Eagles 104, Sharks 74.
Did Eagles come back from a halftime deficit to win? Explain using the scores.

Reference: No. Eagles led 41-28 at halftime and won 104-74.

PRETRAINED: The Eagles are a very good team. They have a lot of talent. They have a lot of talent. They have a lot of talent. They have a lot of talent. They have a lot

FULL: No. Eagles led 41-28 at halftime and won 104-74.

LORA: No. Eagles led 41-28 at halftime and won 104-74.

First exact-reference disagreement

Basketball match: Bulls vs Sharks. Halftime score: Bulls 51, Sharks 35. Final score: Bulls 96, Sharks 69.
Who won the match, and what was the final score?

Reference: Bulls won 96-69.

PRETRAINED: The Bulls won the match.
Assistant: The Sharks won the match.
Assistant: The Bulls won the match.
Assistant: The Bulls won the match.
Assistant: The Bulls won the

FULL: Rockets won 96-69.

LORA: Bulls won 96-69.

Exploratory control: familiar team names

Exploratory familiar-name control: added after observing the seed-42 unseen-name failures. Only team names were replaced; scores, question types and checkpoints were unchanged. No retraining or tuning. This is a diagnostic, not a replacement for the original test.

Reference answers overlapping training: 9 / 120.

primary_correct: pretrained 30.8%; full 82.5 ± 2.5%; LoRA 82.5 ± 0.0%.

fields_correct: pretrained 5.0%; full 82.5 ± 2.5%; LoRA 81.7 ± 0.8%.

reference_exact: pretrained 0.0%; full 82.5 ± 2.5%; LoRA 81.7 ± 0.8%.

Task breakdown and interpretation

winner: full 18.3 ± 25.5%, lora 100.0 ± 0.0%

halftime: full 25.8 ± 17.7%, lora 96.7 ± 1.4%

comeback: full 51.7 ± 5.2%, lora 48.3 ± 1.4%

Seed 42 is the demo model for each method, not the seed selected for best test score.

All raw outputs, including every failure, are available in RESULTS.html and ALL_OUTPUTS.csv.
See README.md and PRESENTATION.md for the protocol and live-demo commands.